# METRO TICKET GENERATING SYSTEM

*Project Planning Phase – Product Backlog, Sprint Planning & Estimation*

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| **Date** | **2 October 2026** |
| **Team ID** | |
| **Project Name** | **Metro Ticket Generating System in ServiceNow Using Service Catalog and Workflows** |
| **Maximum Marks** | **4 Marks** |

**Reference links:** https://www.atlassian.com/agile/project-management | https://www.atlassian.com/agile/tutorials/burndown-charts

## PRODUCT BACKLOG, SPRINT SCHEDULE, AND ESTIMATION

The following product backlog is structured for the Metro Ticket Generating System project using epic-based breakdown, user stories, Fibonacci-based story points, priorities, and assigned team members.

- **Sprint-1 (Epic: Requirement Analysis & Planning):**
  - **USN-1 (Story Points: 3 | Priority: High | Assignee: [Your Name]):** As a commuter, I can submit metro ticket details through a structured Service Catalog Record Producer form.
  - **USN-2 (Story Points: 2 | Priority: High | Assignee: [Your Name]):** As a project team member, I can define the required travel variables, station master data, and user requirements.
  - **USN-3 (Story Points: 3 | Priority: High | Assignee: [Your Name]):** As a project team member, I can design the dynamic Script Include pricing logic and operator workspace workflow.
- **Sprint-2 (Epic: Service Catalog, Pricing & Operator Fulfillment):**
  - **USN-4 (Story Points: 3 | Priority: High | Assignee: [Your Name]):** As a commuter, I can enter origin station, destination station, ticket type, travel class, and smart card ID.
  - **USN-5 (Story Points: 3 | Priority: High | Assignee: [Your Name]):** As a system, I can automatically compute zone-based fares and generate unique digital QR-code pass payloads.
  - **USN-6 (Story Points: 3 | Priority: High | Assignee: [Your Name]):** As a station operator, I can receive catalog tasks and validate active QR passes via the workspace.
  - **USN-7 (Story Points: 2 | Priority: Medium | Assignee: [Your Name]):** As a commuter, I can receive an email notification confirming my ticket booking and QR details.
  - **USN-8 (Story Points: 1 | Priority: Medium | Assignee: [Your Name]):** As a project team member, I can verify the flow execution and document the completed solution.

## PROJECT TRACKER, VELOCITY & BURNDOWN CHART

- **Sprint-1 Tracker:** Total Story Points: 8 | Duration: 5 Days | Start Date: 25 Sep 2026 | Planned End Date: 29 Sep 2026 | Completed Points: 8 | Release Date: 29 Sep 2026.
- **Sprint-2 Tracker:** Total Story Points: 12 | Duration: 5 Days | Start Date: 30 Sep 2026 | Planned End Date: 04 Oct 2026 | Completed Points: 12 | Release Date: 04 Oct 2026.
- **Average Velocity Calculation:** Using a 10-day total planned project duration and 20 total story points across the two sprints, the average team velocity is calculated as:

  *AV = Total Story Points / Total Days = 20 / 10 = 2 story points per day*

- **Burndown Chart Analysis:** The graphical burndown tracking illustrates the steady reduction of remaining work from 20 story points down to 0 points across the 10 planned project iteration days in accordance with standard Scrum methodologies.